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(1) Program / Project Title
Project Title: REQUEST: A Barangay Document Request System for the Residents of West Kamias, Quezon City Overview: It is a centralized web-based system that would provide document requests for Barangay Kamias, Quezon City. It allows the residents to apply for Barangay Certificate and Certificate of Indigency online, track the application status, and get real time notification. Only the barangay chairman, secretary and registered residents can access the system.
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(5) Background and Significance

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Public services are in demand nowadays and they need to be carried out or provided efficiently. Given that Barangay Clearances and Certificates of Residency are traditionally requested using manual processes, they are often inefficient with long waiting times, lost records, and limited resident availability. To tackle these challenges, the researchers are building REQWEST, a centralized web-based system to automate and modernize Barangay Kamias, Quezon City's document request process.

The goal of this system is that residents can easily and quickly apply online with transparency, time indication of their applications, and instant updates or approval notifications. REQWEST minimizes reliance on paper-based transactions, makes processes more efficient, reduces human error, and improves service delivery in the barangay.

This study would be beneficial to the following:

- Barangay Officials – Officials of Barangay West Kamias will benefit from having better access to and management of resident information and document requests. The system will streamline administrative processes, improving efficiency in handling service requests.
- Citizen (Brgy. West Kamais) – Citizens will have easier access to request documents such as birth certificates (PSA), barangay clearances, voter's IDs, barangay IDs, indigency certificates, and business permits. The system also prioritizes certain groups, including:
- Senior Citizens and Persons with Disability (PWD) - These individuals will receive priority service and free service as stated by the barangay secretary.
- Student – Students also receive free services.
- Future Researchers – Future researchers will benefit from this study as it provides a foundation for further development. They can improve and refine the existing system to address emerging needs and enhance its functionality.

(6) Objectives and Research Questions

The main purpose of this study is to develop a centralized web-based system known as REQWEST to enhance the process of requesting barangay documents in Barangay Kamias, Quezon City. The purpose of this study is to develop an online platform for residents to conveniently ask for Barangay Certificates and Certificates of Indigency. It also wants to give a real-time tracking system where users can see their application status and be informed of updates or approval.

One is to digitize barangay operations and make it more efficient by eliminating paperwork and minimizing delays. Additionally, the system will have a document authentication feature that will only process verified requests. Access to the system will be limited to authorized barangay officials and registered residents to maintain the security and security of the system.

This study also intends to see the system's limit as well as possible improvement ways like adding online payment options or using it to other barangays in the future. The study aims to make barangay services more accessible and efficient by introducing this digital solution.

(7) Materials and Methods

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A. Study Design

Definitions - This study focuses on developing a centralized web-based system to process barangay document requests more efficiently. It aims to reduce manual processes and provide a more accessible platform for residents.

Prospective/Retrospective - It is a prospective study as it involves the implementation and evaluation of a new system. The system's impact will be assessed after deployment.

Number of Centers, Randomized? - The study is single-center and conducted in Barangay Kamias, Quezon City. It is non-randomized since all eligible residents can use the system.

Blinded, Placebo, etc. - No blinding or placebo is involved as this study does not include medical interventions. It is purely a technology-based study.

Period of Enrollment or Chart Review Covers? Informed Consent? - The study covers system development, testing, and initial implementation. No informed consent is required as there are no human experiments involved.

Study Drugs, Device, or Intervention - The intervention is the REQWEST system, designed to enhance document processing efficiency. It replaces the traditional manual request system with an online platform.

Instruments - The study will use system logs, standardized surveys, and user feedback to assess system performance and user satisfaction. These tools will help measure efficiency and reliability.

Endpoints: The study aims to measure processing time reduction and improved accessibility. It will evaluate how effectively the system streamlines document requests.

Goals - The goals of the study are to measure the reduction in processing time and the increase in accessibility. It will determine how well the system reduces document requests.

Endpoints - The research project assesses time savings during processing alongside better accessibility levels. The system's capability to simplify document request procedures will be assessed during the study.

B. Study Population

The research includes 100 people residing in Barangay Kamias of Quezon City along with members of the barangay council. The system requires usage from the barangay chairman and secretary together with the scrutinized members of the community. The study includes 100 residents from Barangay Kamias as participants who need either Barangay Certificates or Certificates of Indigency. Internet access capability is necessary for the system users. People who are not residents of Barangay Kamias and those who lack internet access cannot participate in this study. User requests that cannot be handled through system registration must be processed manually by the users. The evaluation lacks experimental or control groups because it investigates document request performance by the system.

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C. Assessment of Resources

Access to the study population

The project needed a comprehensive outreach to both residents and barangay officials from Barangay West Kamias to guarantee success. The survey instrument was made accessible through both digital and physical channels to obtain feedback from numerous participants. The researchers carried out direct interviews with barangay officials while investigating existing challenges and requirements to gather detailed information. The study executed along various data collection techniques to both achieve widespread participant reach and obtain beneficial data that will guide system development.

Time management

The researchers arranged the surveys and interviews in a manner that would be convenient for the participants to increase the possibility of getting more responses. The process of data collection, data analysis, and system design was also segmented in phases to ensure the smooth running of the study. This way, progress reviews were done with the aim of making changes and meeting the set deadlines. This way of time management allowed for the quality of the study to be upheld, and all the main areas to be covered.

Staff training

to enable the barangay officials to use the REQWEST system properly. The researchers also offered instructions on how to use the system, how to deal with documents and how to solve some problems that may arise. The training sessions were quite live, enabling the officials to ask questions and even exercise on the system. Other supplementary items including manuals and instructional guides were also made available for use. It also helped in avoiding any complications as it prepared the company to shift from manual processing to the new system.

D. Study Procedures

Eligible Participants:

- Residents of Barangay Kamias ages 18 and above.

Recruitment Method:

- An online survey form was sent directly to the participants.
- Announcement on the online group community of Barangay Kamias

Survey Study Specifics

- This study employed an online survey using Google Forms and the participants were recruited from an online group of the residents of Barangay Kamias. To achieve these objectives, the survey sought to establish the various challenges that are likely to arise when requesting for documents, the frequency of the request and the need for a web-based system. The survey responses proved useful for the enhancement of the design and features of the system.

Data Collection, Storage, and Disposal

- Open-ended questions were captured using an online Google form and all responses were stored in a Google Sheets that only the researchers could access. Data collected was kept, sorted and used strictly for developing the system. Finally, in compliance with the ethical research, all the participants' details were removed to maintain anonymity.

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Chart Review

- The study also included an assessment of the current process of requesting documents in the barangay. The authors studied the process of handling documents and found out the existing problems with it. From this review, a better system that enhances accessibility, tracking, and efficiency for the residents and the barangay officers was developed.

E. Data Collection

1. Demographic Data

The survey collected essential demographic information, including:

- **Email Address:** Used for verification and follow-ups.
- **Age:** This feature is useful to determine the age of the residents that utilize the system.
- **Address:** Establishing that the participant is living in Barangay Kamias.

2. Personal Reflections

The survey explored the following areas:

- **Requesting Documents Challenges:** Residents mentioned long waiting time, lost paperwork and frequent visit to the barangay office.
- **Survey of Frequency of Document Requests:** The survey found out how frequently residents need barangay documents and peak demand periods.
- **The importance of a centralized Online System:** Respondents indicated that an online system could ease request, cut down the waiting time and increase the efficiency of the service.

Reliability and Validity of the Research Instrument

- **Development** - Survey was conducted using questions related to the challenges that are seen frequently in a document processing environment. It used clear, concise, and unbiased questions in order to obtain the necessary information.
- **Reliability** - The questionnaires were further checked and discussed with the barangay officials to ascertain that it captured the issues on document request and the requirements of the system.

Focus Group Guide

- **Development** - The purpose of the developed focus group guide was to engage the officials of the barangays in discussion regarding the current procedure and possible enhancements.

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- **Facilitator Training** - Participants were orientated on how to conduct the focus group discussions, encourage participants to express themselves and record their inputs.
- **Validity** - The questions were generated with a close focus to the system goals that were to be accomplished to allow the result of the study to be useful in system development.

Sample Size Considerations

The survey involved use of questionnaires to 100 residents in order to capture all the categories of residents in Barangay Kamias. Due to the constraints in timing, the number of participants that were selected in this study was determined by the feasibility of the study and the possibility of getting enough data for data analysis.

- **Power Analysis** - Slovin's Formula was used in this study to come up with the most appropriate number of respondents recommended for the statistical representativeness ratio of the barangay's total population.
- **Justifying the Sampling Procedure** - Online distribution techniques were employed for reaching out to various target groups. This helped in making sure that the sample that was given consisted of the right population in the community.

Statitstical Methodology

- **Convenience between Manual and Online Document Request Processing** – to compare the level of efficiency increase.
- **User Satisfaction Before and After Implementation of the System** – to test the user acceptance of the system.
- **Slovin's Formula:** In this case, the researchers utilized Slovin's Formula in order to come up with the correct number of residents to be surveyed so that the research would be accurate yet feasible to conduct.
- **Categorizing the Data:** The authors mainly collected survey data with open-ended questions and interview feedback responses then coded the data. This was useful in fine-tuning the characteristics of the system and cope with issues highlighted by the residents and the barangay officials.

(8)Safetey and Monitoring Plan

All the gathered data will be stored in a password protected Google Sheets for the sole use of the researchers. To this end, the information to be stored will only be accessible to approved researchers only. The data will be retained for a maximum of six months after the completion of the study and then all records will be disposed off. In this regard, the personal information that may be collected from the patient will not be disclosed to the third party as the patient's information is considered sensitive.

(9) Limitations

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This study only focused on the residents of Barangay Kamias. However, the results may not be generalized to other places with different procedures in document requests. Also, internet dependency may rule out residents who are unable to use the internet or online platforms. The study does not link the online payment transactions or other government systems, but only the barangay level document request automation is done. Barangay officials are still required to give their approval to the application hence, the process is not fully automated.

(10) Ethical Considerations

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A. Informed Consent

The survey respondents were made aware of the nature of the study that was to be conducted and its objectives in advance. Participants' consent was sought by using an option of ticking a box in the Google Forms survey. PWDs and seniors are considered as vulnerable groups and their opinions were prioritized when the data was being collected. The authors made it clear that the participation of participants was on a voluntary basis and that their responses were kept confidential and anonymous. There were no identification data beyond demographics provided, thus ensuring that privacy was not violated.

B. Risks and Side Effects

Based on the activities that will be conducted in the study, the research is equally exempted from any possible harm since it will only entail surveys as well as interviews. Among these, the most frequently mentioned potential inconvenience is the time that the participants spent answering the survey, which was evaluated to be five to ten minutes. In the same manner, there are no risks that are medical, psychological, legal or financial for the participants. There was an option to quit the study regarding its results at any moment without repercussions to the participant.

C. Benefits to Subjects

This means that the residents will be able to request and access documents within a relatively shorter period of time. This will also enhance the workflow in the barangay since some work will be minimized to be done manually. This may help in the enhancement of the future digital governance within the barangay.

D. Costs to Subjects

There were no monetary expenses involved in participant participation. The online survey format needed only access to the internet which the majority of residents already possessed as part of their regular activities. The research process specifically focused on accommodating PWDs and seniors to prevent any disadvantage when they participated.

E. Compensation to Subjects

The research did not award any monetary payment to study participants. All subjects could participate by free choice without receiving any rewards.

F. Provisions for Vulnerable Subjects

- A special data collection process prioritized PWDs and seniors as the primary subjects to guarantee their issues received attention.
- For subjects needing help, special arrangements included survey assistance.
- The surveyed population received easy-to-understand designs in their questionnaires to support understanding.

G. Subject Privacy and Data Confidentiality

Privacy of Participants

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- The collection of sensitive personal information stopped after obtaining email addresses together with age and residential address.
- Specific measures ensured complete privacy protection for personal information belonging to PWDs and seniors.

Confidentiality of Data

- The Google Sheets database stored data behind a protected password.
- The research data remained accessible only to authorized investigators.
- The research findings were displayed through aggregate results without showing any specific participant records.
- All research-related information remains strictly limited within the team conducting the study.
- The study data will be kept safely for six months following the completion of the research project.
- The study records will face permanent deletion after the designated period ends.

Limits to Confidentiality

- The research provides confidentiality protection but does not have the ability to restrict participants from revealing their personal information.
- The online survey data collected through Google operated under the privacy regulations set by Google.

(11) Plan for Dissemination of Findings

Research findings will be presented at the institution where the capstone project takes place during defense sessions. The final results will be presented to Barangay Kamias officials to enhance their document request system. Academic paper presentations and local research conferences may accept the study for submission if these findings correspond to the subject of barangay digitalization.

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(13) Appendices

This refers to research instruments, consent forms, communication letters etc. are submitted along with this paper.

Signature	(14) Prepared by :	(15) Endorsed by:
		
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